

IILM UNIVERSITY

GREATER NOIDA, UTTAR PRADESH

AICTE – EDUSKILLS VIRTUAL INTERNSHIP

Advanced Robotics Using AI and IoT

8-Week Virtual Internship Presentation

STUDENT DETAILS

Anurag Betal

Roll No.: 2410030182

B.Tech (Computer Science & Engineering)

EduSkills Academy

June 2026 – August 2026

Candidate & Program Overview

Student Name

Anurag Betal

Roll Number

2410030182

Degree

B.Tech (Computer Science & Engineering)

Academic Batch

2026–2027

Program

AICTE – EduSkills Virtual Internship

Specialization

Advanced Robotics Using AI and IoT

Organization Profile – EduSkills Academy

- EduSkills Academy focuses on industry-oriented learning, skill development and employability.
- The organization provides virtual internships and training opportunities in emerging technologies.
- Programs cover areas such as Artificial Intelligence, IoT, Robotics, Cloud Computing and Cybersecurity.
- The internship model helps students connect academic concepts with practical, technology-focused learning.

Problem Statement

Challenge

Traditional robotic systems can rely on predefined instructions and limited sensing, reducing their ability to respond to changing environments.

Need

Modern robots should sense their surroundings, process information, make suitable decisions and communicate system information remotely.

Project Focus

Explore the integration of Robotics, Artificial Intelligence and IoT to create smarter, connected and more autonomous robotic applications.

Project Objectives

Robotics Fundamentals

Understand sensors, actuators, motors, controllers and robot operation.

AI Integration

Explore intelligent decision-making and autonomous robotic behavior.

IoT Connectivity

Understand wireless communication, remote monitoring and data exchange.

System Control

Process sensor inputs and convert decisions into suitable robot actions.

Practical Skills

Build knowledge of programming, embedded systems and troubleshooting.

Technologies & Tools Used

Robotics

Sensors, actuators, motors, motor drivers and robotic mechanisms.

AI

AI/ML concepts for perception and decision-making.

Microcontrollers

Arduino, ESP32 or ESP8266 for control and sensor interfacing.

Programming

Python and C/C++ for logic, control and automation.

IoT

Wi-Fi, Bluetooth, MQTT and HTTP for communication.

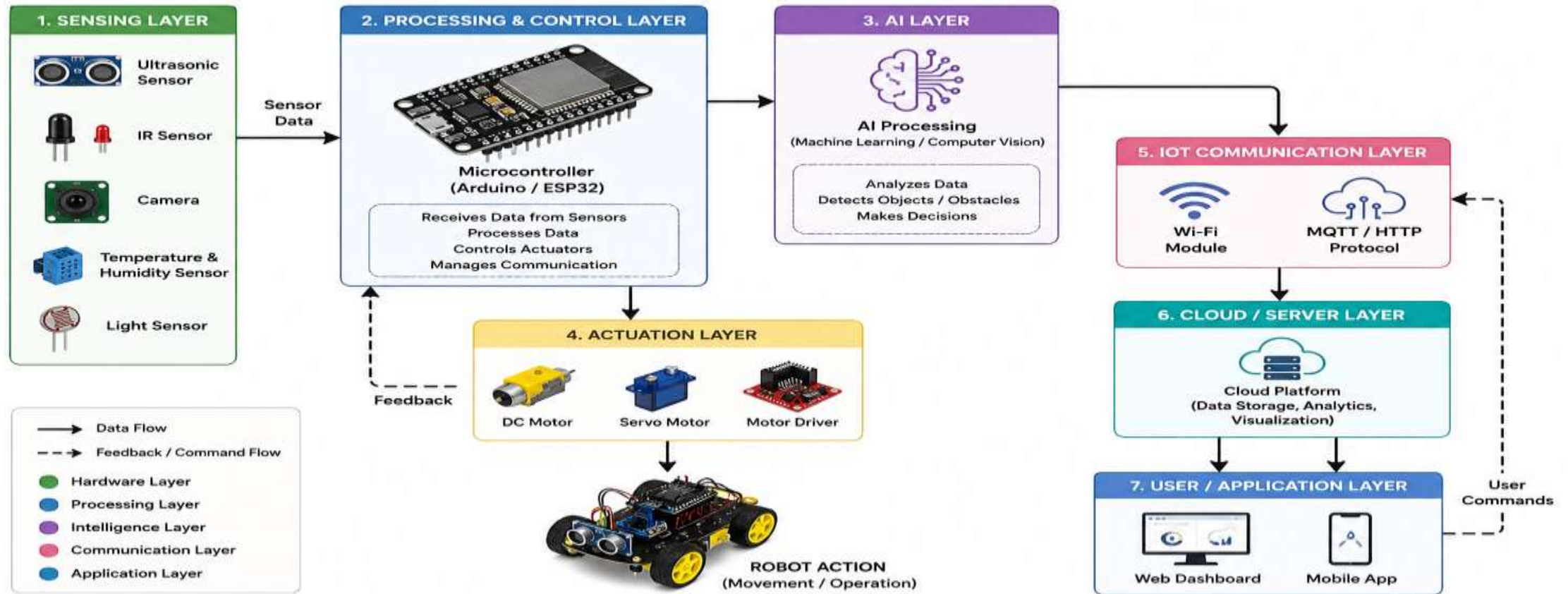
Cloud / Monitoring

IoT or cloud platforms for storage, visualization and remote monitoring.

System Architecture

Sensors → Controller → AI Decision → Actuators, with IoT connectivity to cloud and user applications

SYSTEM ARCHITECTURE Advanced Robotics Using AI and IoT



Methodology

1. Requirement Analysis

Identify tasks, sensors, processing and communication needs.

2. System Design

Define sensing, processing, AI, actuation and IoT components.

3. Data Collection

Collect real-time information from sensors.

4. Data Processing

Process inputs through a controller or computing platform.

5. AI Decision

Analyze information and determine an appropriate action.

6. Actuation

Control motors or other actuators.

7. IoT Integration

Transmit data for remote monitoring and communication.

8. Testing

Evaluate accuracy, response, reliability and overall operation.

Internship Learning Timeline

Weeks 1–2

Robotics Fundamentals

Sensors, actuators, motors, controllers and basic robot operation.

Weeks 3–4

AI for Robotics

Perception, obstacle detection, decision-making and autonomous behavior.

Weeks 5–6

IoT Integration

Wireless communication, data transmission and remote monitoring.

Weeks 7–8

Integration & Evaluation

System integration, testing, optimization and project documentation.

Expected Outcomes

- Understand the fundamental architecture and operation of robotic systems.
- Gain knowledge of AI-based decision-making and autonomous robotic behavior.
- Understand IoT communication, remote monitoring and connected robotic applications.
- Develop practical awareness of sensors, actuators, microcontrollers and programming.
- Improve troubleshooting, system analysis and problem-solving skills.
- Build a foundation for future work in automation, smart agriculture, healthcare, surveillance and autonomous systems.

Scope of the Project

Smart Robotics

Integration of sensors, controllers, AI and actuators for intelligent robotic operation.

Connected Systems

IoT-based communication, remote monitoring and data exchange.

Automation

Reduced human intervention through sensor-based and AI-assisted control.

Applications

Industrial automation, agriculture, healthcare, surveillance, logistics and smart environments.

Internship Completion

Student

Anurag Betal

Roll Number

2410030182

Organization

EduSkills Academy

Completion

Successfully completed the 8-week Advanced Robotics Using AI and IoT Virtual Internship.

THANK YOU

Anurag Betal

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Advanced Robotics Using AI and IoT